NASA SUITS 2026 Proposal: Team Tuffy Space

NASA Spacesuit User Interface Technologies for Students (SUITS), 2025–2026
California State University, Fullerton
Public version of our team's submitted proposal. Contact details, signatures, and the NASA administrative template have been removed. Submitted October 30, 2025.

Team

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1. Technical Section

1.1 Abstract

Our team proposes a system that integrates crew vitals, navigation, and AI-assisted decision support into a unified AR interface for lunar EVA operations. A persistent panel displays critical biomedical data such as heart rate, oxygen levels, and suit power using color-coded indicators that remain visible under all configurations. A 2D topographic map anchored to the left side provides real-time tracking of the EV crew, surface assets, pressurized rover (PR), and LTV, showing traversed and planned paths, hazards, and user-defined pins. The interface processes LTV beacon signals to estimate its position, continuously refining the search area and generating optimal, breadcrumb-marked routes for both out-of-bound travel and safe return. Autonomous and perception systems, using LiDAR, odometry, and visual SLAM, support hazard detection, localization, and dynamic rerouting. Alerts appear in proportion to their severity,

paired with AI Assistant (AIA) recommendations. The **System Status** display summarizes rover power, thermal and life support systems, and battery life, while mission timer logic ensures oxygen safety by preventing EVA commencement if reserves are determined to be too low. Simple task workflows minimize cognitive load through interactive modules: **Show Biometrics** reveals health data with threshold-based prompts; **Show UIA** presents a guided pre-egress checklist, supported by AIA which will guide you through each step; **Show ERM Suggestions** assists with LTV diagnostics and repairs; and **Show Ingress Steps** leads the astronaut back to the rover along verified paths while monitoring for new hazards and scanning for more efficient pathways.

Usability and performance will be tested through a Human-in-the-Loop campaign that progresses from peer and faculty reviews to iterative user testing and final optimization. Evaluation metrics include task completion time, navigation efficiency, alert comprehension, and error rates under simulated telemetry and beacon conditions. Field validation will use a HoloLens 2 running the full application with simulated inputs for vitals, rover systems, and LTV beacons, with emulated sensors as needed. This prototype directly supports NASA SUITS 2026 objectives by enhancing crew safety, navigation, and repair operations while maintaining the astronaut's focus on the lunar terrain ahead and current tasks at hand.

1.2 Software and Hardware Design Description

1.2.1 Pressurized Rover (PR) Interface

UI Display. The UI shall display a 'Crew Vitals' status panel in the upper-right corner for all active EVs. This panel shall be expandable and collapsible via voice command. The 'Crew Vitals' panel shall include, at a minimum: Heart Rate, O2 Levels (%), Expected Oxygen Life (%), and Suit Battery Level (%). The display shall be customizable by the user to show preferred parameters and layouts, and can be expanded via voice command. Critical biomedical data shall remain displayed regardless of user configuration. The system shall display vital sign data from real-time telemetric streams. Vital signs shall be represented using icons and color-coded indicators to differentiate between nominal (expected) and off-nominal (abnormal) states.

Navigation/Map Interface. Positioned left-most, a 2D map displays the live position of surface assets, pressurized rover, LTV, and the EV crew members including their expected route and traversed route for each mission. The map displays projected and traversed paths of EVA crew members, displays mission-defined points of interest, and pins that can be placed by the user to caution for hazardous conditions and/or obstacles in the way. Additionally, pins added by the

EVs can be seen on the map in a different color.

LTV beacon data is displayed on the map with directional prompting such as direction and distance. New LTV beacon data is fed through the **data handler** (located in the bottom left) and into the map, which dynamically updates the new potential travel radius of the LTV. Longer intervals between each beacon data update lower how accurate the travel radius of the LTV is; in parallel, the signal strength color will begin to lighten, indicating poorer tracking accuracy. The data handler shows when the last beacon data was sent. The map will display the anticipated search route, which updates dynamically based on the potential travel radius.

Warning/Caution Notification System. The system shall interrupt and alert in the HUD; depending on the severity of the warning or caution, an appropriate interruption shall be sent. Warnings will be sent when telemetry values are off-nominal, such as vitals dropping below an acceptable operation level, when a caution event has been triggered, and when an obstacle is currently being avoided. For each warning or caution, the system shall display the AIA-recommended procedure.

System Status. The System Status display shall show the status of all designated systems. Designated systems shall include, at a minimum: rover battery, vehicle wear, cooling system temperatures, internal temperature, and fan speed. The system shall use AIA to provide status updates in simple, neutral language and shall use predictive models to estimate remaining battery life.

Mission Timer. A Mission Timer shall be displayed in the format HH:MM:SS showing the elapsed time since mission start; expected mission time will be displayed in the same format. The system shall not permit a mission to start if the active EV's oxygen level is projected to be below 30% at the expected mission completion time. All mission timers will adhere to standards and constraints given to our team.

Autonomous Navigation System. The system uses beacon signal strength (RSSI) to estimate the direction and distance to the LTV. Direction is then triangulated using multiple readings as the rover moves, which gives a position estimate of the LTV that is placed on the map as a "target waypoint". The path planner then creates an obstacle-free route to the target waypoint. As the rover moves and the beacon data updates, the route will be changed dynamically. LiDAR and depth sensors detect obstacles and work with the path planner to move toward the LTV while avoiding obstacles to maintain safety. The obstacle detection model works with Mission Control, which adds any hazard zones it sets on the map as obstacles to be avoided. Odometry, IMU, and visual SLAM are used to track speed and heading.

Outside View. The Outside View can be toggled between three different views: Default,

Pass-through LiDAR view, and LiDAR view. In any LiDAR mode, all obstacles shall be highlighted along with danger zones sent from Mission Control. Waypoints and/or points of interest shall be highlighted as a ray beam rising from the ground like a beacon, easy to see from a distance. Waypoint visibility shall be configured based on distance, strength, and color, and can be toggled off.

1.2.2 EV Spacesuit Design

The EV spacesuit system is designed to be intuitive and effective, allowing users to accomplish desired tasks efficiently while not obstructing their vision or cognitive abilities. This will be accomplished by grouping information into sections and making each section modular, allowing the EV to customize what information they are being presented with at all times.

2-Dimensional map. The 2-Dimensional map will rest in the top left corner of the HUD and can be expanded using voice commands or hand gestures if the EV wishes to have a more focused view of the map. The map will be top-down and topographical, centered around the EV with a 50 meter radius. The color of the map will vary with depth to make it easy to see where nearby craters and boulders are. The map will also feature a recommended path to take based on the determined or predicted position of the LTV and detected obstacles. The path will be displayed as a pink line leading to the LTV's last known location from the position of the EV. There will also be a dotted pink line trailing the EV leading to the PR, allowing the EV to follow it back to the PR as a **breadcrumb** feature. The map will also have colored dots that display the real-time locations of other EVs. The dots will be colored based on the EV's health status: green being healthy, yellow meaning monitor more closely, and red meaning one or more areas of that EV's biomedical data is coming back as compromised. The map will also display any **pins** the EV wishes to drop themselves and pins the software automatically drops. There are three different pins the EV can choose from:

- **Danger Pin:** Appears as a red circle and will be automatically placed over dramatic drops or increases in depth to denote potential obstacles. They may also be manually placed.
- **Custom Pin:** Appears as an orange square and can only be created by the EV. The EV will have the option to label the pin upon creation, and labels can be edited after the pin is already dropped.
- **Navigation Pin:** Appears as a pink triangle and can only be placed by the EV. These pins will automatically generate a best-travelled path between them and the EV and override the default path set when there are no pins.

These pins can be placed in two ways. The first is by expanding the miniature map and then

pointing to a location on the map. The second is by pointing at a location you are looking at and then using voice commands to instruct the HUD to "drop (pin type) pin". You may also command the system to "Navigate to pin" by first selecting the pin you wish to travel to, then speaking the command. Any pin you set a navigation to will have the best path drawn to it from the user. Lastly, the map will automatically highlight a recommended area to search for the LTV, visualized by a transparent yellow circle on the map. This area will be determined using a search algorithm that predicts a general area the LTV could be in based on the obstacles in the area and the previous path it was taking.

Spacesuit Telemetry. The spacesuit will also feature a **biomedical data** module. The EV can call upon this module by speaking the command "Show Biometrics". This will bring up a transparent module that displays the EV's current heart rate, body temperature, blood pressure, and breath rate. If any of these categories fall under a specified threshold, the HUD will display a warning logo and a discreet prompt, comparable to a phone notification, near the minimap in the top left corner of the HUD asking the user to review biomedical data. Upon opening the warning, all sections of biomedical data that are below the threshold will be highlighted and a recommendation for mitigating steps will be generated using AI that evaluates how many problem areas there are and what needs to be done to assist each area.

Procedure List. Our HUD will also feature a program to check that the EV has completed all checklist items in order to safely **egress**. When speaking the command "Show UIA", the HUD will show a digital version of the UIA with each port and switch outlined in red. The red outline shows the EV that they have not completed that step yet or that something is wrong with the connection. Adjacent to the digital UIA will be a checklist to walk the EV through each step they need to complete before initiating a safe egress. For more detailed audible instructions, the EV can speak the command "show egress step" followed by the step number; that command will begin an audio recording of instructions on how to complete that step. Once each necessary piece of the checklist is complete, the corresponding port or switch on the digital UIA will glow green to show the EV that it is prepared and ready. Before exiting the PR, the 2-Dimensional map will be updated with a yellow ring that displays the range of the life support systems. While outside the rover, the HUD will show a range converted into a percentage that tells the EV when to consider turning back. At around 65%, the HUD will notify the EV that they should turn back.

ERM. Our system will also make use of AI for recovering and repairing the rover. Upon reaching the LTV, the EV can use the voice-activated AI assistant to scan the LTV for damage or complications. The EV says "Show ERM suggestions" to activate the scanning procedure. The EV looks at the LTV one side at a time, and the AI assistant outputs written suggestions into a

notes section on how to extract the rover based on the obstacles and the reason the LTV is out of commission. Once extracted, the AI assistant gives further detail on the status of the LTV and detailed audible instructions on how to reboot the LTV's navigation based on a diagnostic run on the system. Once the reboot is complete, the AI assistant reruns the diagnostics and confirms that the position data is correct and the navigation system is ready to continue operating. Lastly, before initiating ingress, the AI does a final full 360° scan of the physical components attached to the LTV. Based on the scan, the AI assistant suggests any physical repairs that need to be made to sensors or critical components such as wheels. Once the AI has done a final check and confirmed all physical systems are working, it advises the EV to start the LTV and initiate ingress.

Ingress. Ingress will initiate when the EV has completed rebooting the LTV. However, an early ingress will be initiated in case of critical warning signs such as low oxygen levels or concerning biomedical data. To initiate ingress, the user speaks the command "Show ingress steps", which activates the AI assistant to assist the EV in navigating back to the rover. The EV follows the **breadcrumbs**, displayed as a dotted line trailing the EV on the miniature map and as a dotted line trailing the user on the ground through the HUD. The AI continues to monitor the path as the EV follows the breadcrumbs to ensure it has not been compromised. Once successfully back inside the rover, the AI assistant once again displays a digital UIA and guides the EV through the steps of disconnecting from it with audible instructions.

Necessary Hardware. The HoloLens 2 program will be sufficient to provide all additional hardware we need.

1.3 Concept of Operations (CONOPS)

1.3.1 Rover Overview

Our design for the Pressurized Rover (PR) is based on several operational assumptions. First, all rover interfaces and augmented visual elements are assumed to be integrated into the PR's onboard display and fully compatible with the EVA systems and communication networks. Second, the rover maintains an uninterrupted telemetry link with Mission Control, the Lunar Terrain Vehicle (LTV), and EVA crew members to allow for continuous data sharing and live positional updates. Third, the rover's autonomous navigation and sensor systems are fully operational, including LiDAR, depth cameras, odometry, and IMU inputs, which enable obstacle detection and mapping. Finally, it is assumed that the rover's onboard AI assistant (AIA) is active throughout the mission to provide real-time updates, warnings, and recommended

procedures to support safe operations.
The Pressurized Rover is designed to serve as a mobile operations hub for EVA missions, providing both crew and mission control with real-time situational awareness, vehicle health monitoring, and semi-autonomous navigation capabilities. The system consolidates telemetry, navigation, and status indicators into a single intuitive interface, allowing the rover crew to coordinate effectively with EVA members and the LTV. By combining real-time telemetry visualization, dynamic mapping, and AI-assisted decision support, the PR ensures mission safety and efficiency while reducing crew workload and maintaining continuous coordination with EVA operations.

1.3.2 Spacesuit Overview

In order for our design to function properly with the spacesuit, we run under the assumption that the HoloLens 2 is functional within the system of the suit. We also assume that all biomedical sensors can be implemented into the spacesuit to provide the required data for our biomedical monitor. Our team proposes a design under three core characteristics: intuitive, personalization, and safety.
We designed each aspect of the suit to work intuitively, with two ways to interact with the spacesuit interface: voice commands and hand gestures. The combination of these two allows the user to accomplish any goal they have on the interface by simply telling the suit what they want to see and then interacting with it through hand movements. Every task is also made intuitive through AI assistance. For example, in ERM the AI assistant reduces cognitive load and makes the repair process more intuitive by highlighting problem areas and determining the status of the LTV.
We achieve personalization through our modular UI. Every digital component of the UI can be closed and stored completely out of sight through the voice command system. When you want to bring up a specific system, you speak the appropriate command and it comes into view in a non-intrusive way. If you want the component to take up a large portion of your FOV, you can interact with it using hand gestures and zoom in or out.
We accomplish safety by implementing AI into the egress and navigation systems. The AI assistant is capable of walking you through every step of the pre-egress checklist. It highlights areas on a digital UIA that you have not completed yet and provides written and audible instructions for the EV to follow. Safety is also achieved through AI integration into range recommendations and biomedical monitoring. When outside the rover, the EV has access to recommendations from the AI on how long they can remain outside. If any health section

becomes compromised, the AI assistant gives audible suggestions on mitigating steps to ensure the survival and health of the EV. Finally, safety is achieved in both egress and ingress through the AI-assisted navigation system, which scans the EV's field of view, notes any obstacles, and chooses the safest and most effective path accordingly.

1.4 Human-in-the-Loop (HITL) Testing

Testing schedule (5-month development period):

- **November:** Reserved for development. Outreach for AR demo offers will be sent via email or other methods in the case of preexisting contacts. Before demos occur, peer and professor review will be documented and recorded as the preliminary testing phase.
- **December:** 2–3 early demos, iterating on both the demo process and any major functional bugs and errors encountered. Demos include a short "Mission Brief" of the task scenarios, followed by as many volunteer tests as time allows.
- **January:** 2–3 formal demos with an improved demo process, allowing greater focus on qualitative observations of the user experience.
- **February:** 2–3 formal demos to hone in on what remains to be improved.
- **March:** 2–3 polished demos, which will not result in major changes to the systems (unless absolutely necessary), instead gathering data about quality-of-life changes.

1.5 Project Management

We plan to follow a structured timeline spanning from January 2026 to April/May 2026, splitting the workload of developing the EV HMD system and the PR system in tandem between the team to speed up development. At regular intervals, we plan to test and ensure compatibility between each system while in development, spanning from initial setup, main development, and final testing/integration with human-in-the-loop tests to ensure we meet the core values of our project.

2. Outreach (Community and Industry) Engagement Section

2.1 Community Engagement

Objectives:

1. Get a diverse set of data for Human-in-the-Loop testing of our UIs.

2. Introduce the next generation to XR tools, and provide guidance to further interest in the technology.

Plan:

- Reach out to local high school clubs that may be interested in an AR demo (e.g., Space Exploration Club, Video Game / Game Dev Club, Engineering / Computer Science Clubs).
- During the demo, provide a presentation with a mock "mission brief" explaining the tasks the demo volunteers will complete. Document any found bugs, positive/negative reception, difficulties, and other features of note.
- The number of demo sessions may vary depending on responses from the groups reached out to.

2.2 Industry Engagement

Potential industry partners:

- **Unreal Engine (Epic Games):** Partnering with this company gives us the highest chance of success in our outreach plans because our UIs revolve around their engine, a great opportunity to show the potential of their product. This engagement would also pair well with our high school demos, because the younger audience is already familiar with products, specifically video games, made with their engine.
- **Microsoft:** A strong option to collaborate with because their HoloLens 2 is vital to our development cycle and community engagement. With the resources available to them, it may be feasible to request mentoring or other forms of help.
- **Local professors:** Reaching out to local professors gives us a high chance of a meaningful response or insight. They may be willing to direct us to useful connections, including fellow professors, colleges, previous students, or industry contacts.
- **Local startups:** Although local startups may not have many resources, correspondence is likely in the mutual interest of gaining connections and sharing our projects. This exchange may be particularly useful for understanding how a startup navigated modern challenges with a small team and limited resources.

3. Budget

Item	Cost
Transportation	$50
Outreach events (food)	$100
HoloLens 2 purchase (in event of no loan)	$4,000
Total	**$4,150**

A loaned HoloLens 2 from the NASA SUITS program was requested for participation.